Emerging Methods of Auction Design in Social Networks

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Abstract

A new scheme of auctions: Networked Diffusion Auction

- Process: (1): market spontaneous expansion; (2): allocation and pricing implementation.
- Benefits: More potential participants: **higher social welfare** and **higher revenue**.
- Challenges: Conflicts: auctioneer(more players brings payoff) and bidders(tougher competition).
- Ideas: Explicit each bidder's personal competitors; define reasonable diffusion reward.

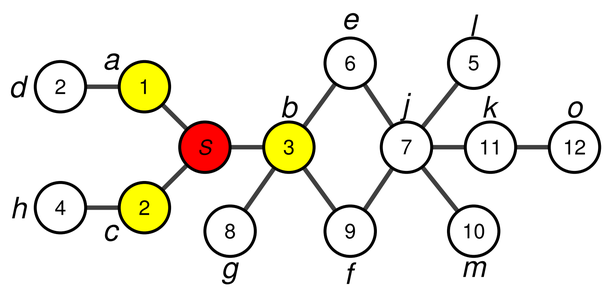
Networked Auctions Overview

Networked Auctions	Single-Item Auctions	Unweighted Graph	Diffusion-VCG [Li *et al.* 2017]
			Information Diffusion Mechanism [Li *et al.* 2017]
			Critical Diffusion Mechanism [Li *et al.* 2019]
			Fair Diffusion Mechanism [Zhang *et al.* 2020]
			Network-based Redistribution Mechanism [Zhang *et al.* 2020]
			Aligned Path Mechanism [Takanashi *et al.* 2019]
			Fixed Price Diffusion Mechanism [Zhang *et al.* 2019]
			Groupwise-Pivotal Referral Mechanism [Jeong and Lee. 2020]
		Graph with Intermediaries	Customer Sharing Mechanism [Li *et al.* 2018]
			Single/Multi-Level Diffusion Mechanism [Li *et al.* 2020]
		General Weighted Graph	Weighted Diffusion Mechanism [Li *et al.* 2019]
	Multi-Items Auctions	Multi-Unit Auctions Single-Unit Demand	Distanced-based Network Auction Multi-Unit demand [Kawasaki *et al.* 2020]
			General Information Diffusion Mechanism [Zhao *et al.* 2018]
		Double Auctions Multi-sellers/buyers	Double Networked Auction Mechanism [Xu *et al.* 2019]
		Procurement auctions With Budget Limit	Budget-feasible Diffusion Mechanism-Homogeneous/Heterogeneous Items [Liu *et al.* 2021]
		Multi-Unit Auctions Down-Sloping Valuations	General Aligned Path Mechanism [Takanashi *et al.* 2019]

Single-item Auction On Unweighted Graphs

Representative diffusion mechanisms

$\mathcal{M}$	Allocation: π	Payment: $\mathbf{x}$
VCG	$SW^*(\mathbf{t}')$	$SW(\mathbf{t}'_{-i}) - (SW(\mathbf{t}') - \pi_i v_i)$
IDM	$SW^*(\mathbf{t}'_{-(i+1)})$	$SW(\mathbf{t}'_{-i}) - (SW(\mathbf{t}'_{-(i+1)}) - \pi_i v_i)$
CDM	$SW^*(\mathbf{t}'_{-\alpha_i})$	$SW(\mathbf{t}'_{-i}) - (SW(\mathbf{t}'_{-\alpha_i}) - \pi_i v_i)$
FDM	$SW^*(\mathbf{t}'_{-(i+1)\cup M_{i,i+1}})$	$SW(\mathbf{t}'_{-i}) - (SW(\mathbf{t}'_{-(i+1)\cup M_{i,i+1}}) - \pi_i v_i) - R_i$
NRM	$SW^*(\mathbf{t}'_{-T(i)})$	$SW(\mathbf{t}'_{-i}) - (SW(\mathbf{t}'_{-T(i)}) - \pi_i v_i) - R'_i$



Numerical Results in the Example

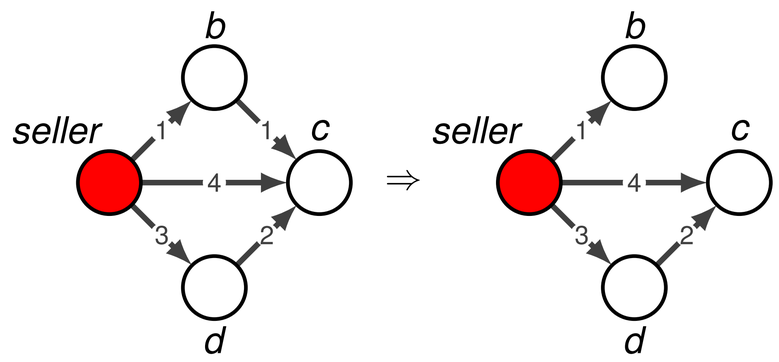
$\mathcal{M}$	Winner	Rewarded bidders	SW	$Rev^{\mathcal{M}}$	Efficient
SPA	$b(2)$	$\emptyset$	3	2	✗
VCG	$o(11)$	$b(-8), j(-3), k(-2)$	12	-2	✓
IDM	$k(10)$	$b(-5), j(-1)$	11	4	✗
CDM[a]	$k(10)$	$b(-4), j(-1)$	11	5	✗
FDM	$k(10)$	$b(-4), e(-\frac{1}{3}), j(-1)$	11	$\frac{14}{3}$	✗
NRM	$k(10)$	$a(-\frac{8}{13}), b(-\frac{18}{13}), c(-\frac{4}{13}), e(-\frac{5}{8})$ $f(-\frac{1}{2}), g(-\frac{5}{8}), j(-\frac{5}{2}), l(-\frac{1}{4})$	11	$\frac{61}{26}$	✗

- **Second Price Auction(SPA)**: low efficiency and revenue in limited market;
- **Diffusion-VCG**: efficient but not weakly budget balance (WBB);
- **IDM**: IR, IC and WBB but not efficient;
- **CDM**: diffusion mechanism family;
- **FDM**: consider fair reward for nodes;
- **NRM**: redistribute revenue in social network.

[a] Here take α_i as the minimum cutting edges subset blocking next critical diffusion node.

Single-item Auction On Weighted Graphs

- Allocation rule considers transferring cost.
- Non-critical diffusion nodes changes the allocation results!
- Critical diffusion sequence does not work!

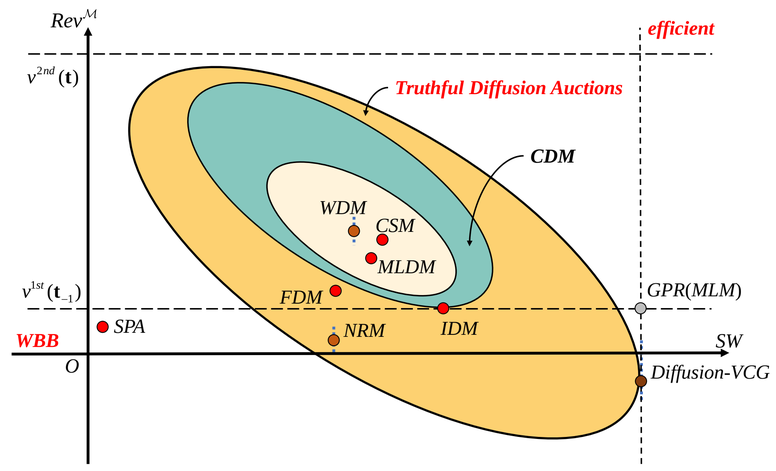


If b does not diffuse information to c, results may totally changed!

- General Weighted Graphs: **WDM**
  - **Allocation**:First bidder satisfying $v_i = v^*_{-\gamma_i}$ in min-cost trading path.
  - **Payment**: Winner: Critical bid beating all on-path nodes before her; Loser on path: Social welfare decrease correlated with their γ_i.
- Graphs with intermediaries: **CSM**

  Encourage merchants to share customers!
  - **Allocation**: The agent whose trading path would maximize the social welfare.
  - **Payment**: Winner: VCG payment; Intermediaries: Social welfare decrease correlated with their threshold neighbourhood set.

Single-item Diffusion Auction Research Map



- Yellow region: all truthful diffusion auction. Green region: critical diffusion mechanism family.
- **Diffusion-VCG**: IR,IC,efficient but not WBB.
- **IDM**: IR, IC, WBB, lowest revenue bound in CDM.
- **GPR(MLM)**: IR, efficient, WBB but not IC.

Single-item IR & IC Diffusion Auctions

- ✓ Value-monotonic allocation rule.
- ✓ Winning/losing payment: bid-independent, diffusion-monotonic.
- ✓ Critical winning bid equals winning payment minus losing payment under truthful diffusion.
- ✓ No diffusion losing payment never goes beyond zero.

Multi-items Diffusion Auction

Different Scenarios:

- **GIDM & DNA-MU**: Multi-unit diffusion auction with single-unit demand.
- **GAPM**: Multi-unit diffusion auction with down-sloping valuation bidders.
- **DNAM**: Networked auction in independent double auction markets.
- **BDM-H/H**: Networked procurment auction with budget.

Diffusion Incentive Design

- **Crowdsouring**: DARPA 2009 red balloon challenge; data acquisition via social networks;
- **Matching**: Modified TTC under acyclic preference and tree network;
- **Information Propagation** with budget limit;
- **Influence Maximization** in social networks.

Open Questions

- **New Models of Diffusion Auction**:
  - ✏ Optimal networked auction design.
  - ✏ Networked combinatorial auctions?
- **Properties of Diffusion Auction**:
  - ✏ Robustness of diffusion auctions: sybil-proofness and collusion-proofness?
  - ✏ Different networks influence result of mechanisms?
- **Other Diffusion Incentive Design**:
  - ✏ Voting: Reducing cost under networked voting while guaranteeing efficiency?
  - ✏ Rating: Extending personalised rating systems with more agents?
  - ✏ Any other scenarios need attracting more participants?